

FINTECH & COMMERCE · IQOO HACKATHON 2026 · PUNE
CITY BATTLE SUBMISSION

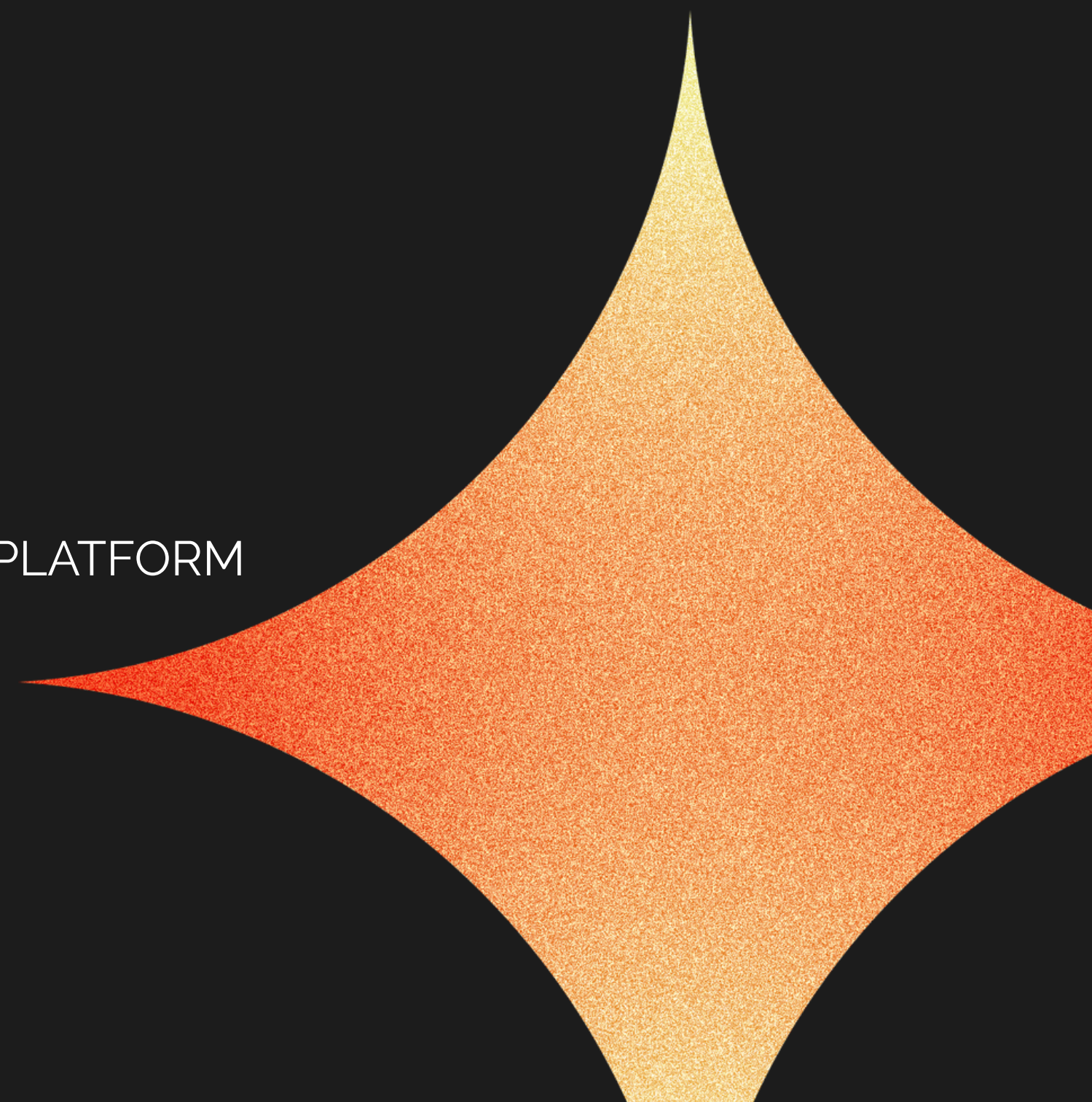
ARTHIX

SHAKE & AUDIT

ZERO-FRICTION, PHONE-FIRST INTELLIGENT PERSONAL FINANCE PLATFORM
WITH MULTIMODAL CAPTURE & GROUNDED ON-DEVICE AI

TEAM NAME :**OLIVE OIL**
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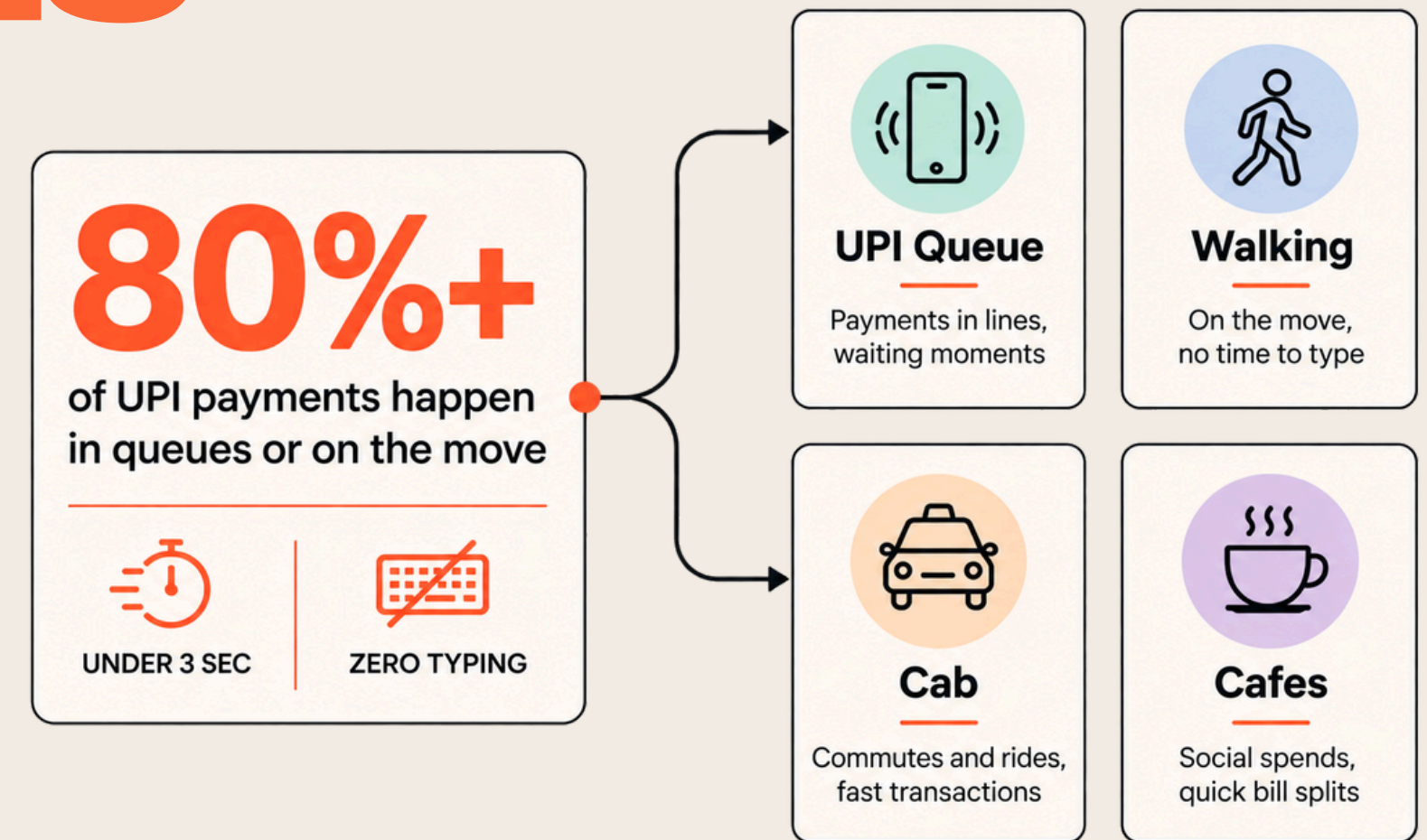
CONTENT:
WHY
FOR WHO
HOW



WHY EXPENSE TRACKING FAILS

THE 3-SECOND FRICTION GAP AT CHECKOUT

- **Payment happens in seconds:** users are in queues, walking, cabs, or meetings with hands occupied.
- **Manual capture is the real friction:** existing AI trackers still require users to enter the data first.
- **Missed logs compound daily:** incomplete records create an unreliable picture of spending.
- **Result:** 80%+ of micro-transactions go unlogged → zero financial awareness.



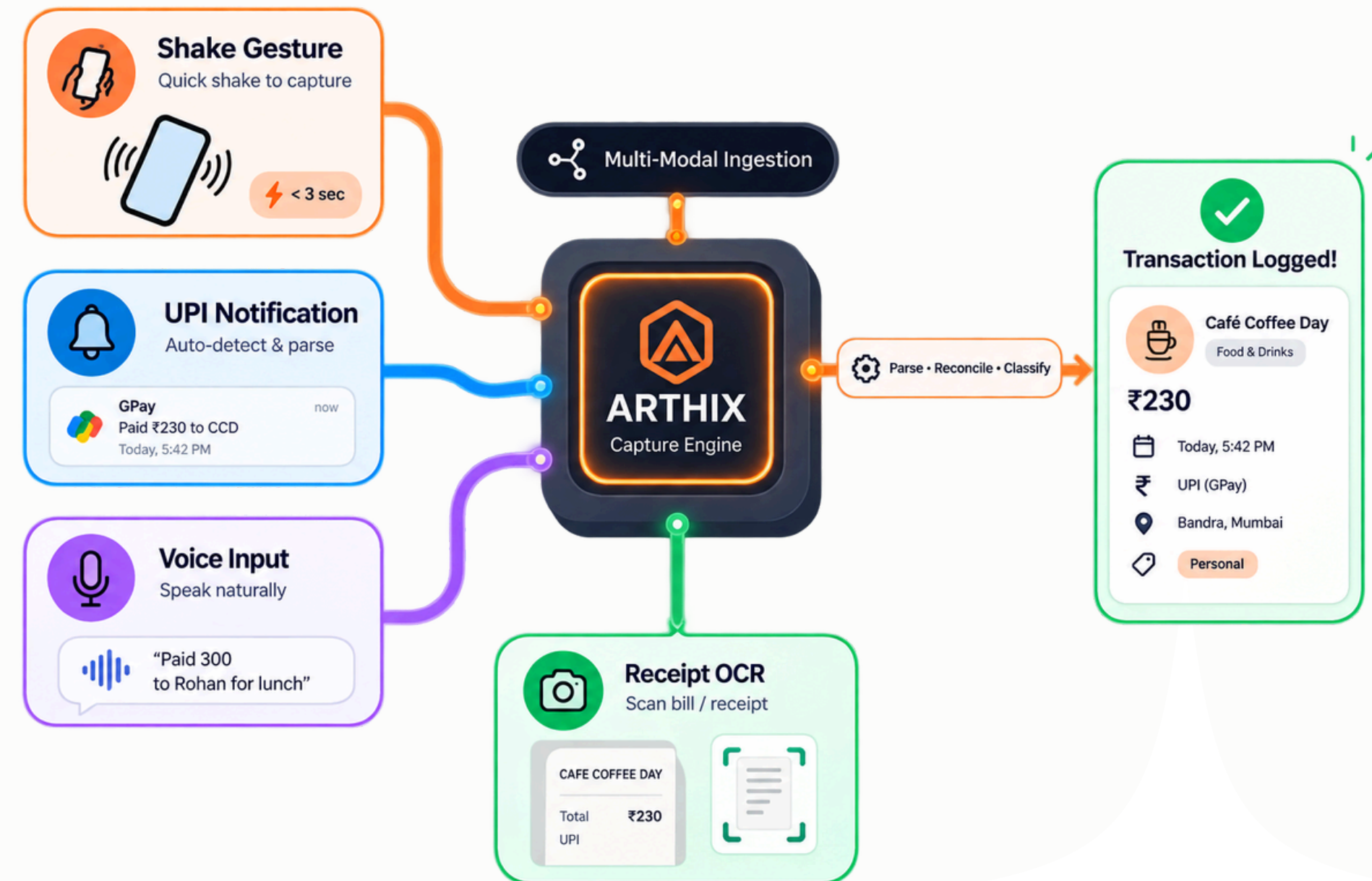
THE PROBLEM ISN'T THE APP. IT'S THE 3 SECONDS YOU DON'T HAVE.

MULTIMODAL, GESTURE-FIRST, AUTONOMOUS EXPENSE INTELLIGENCE



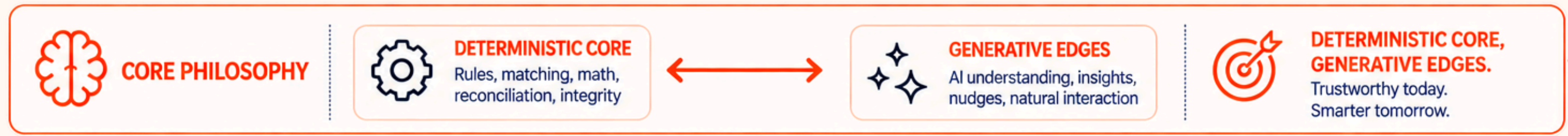
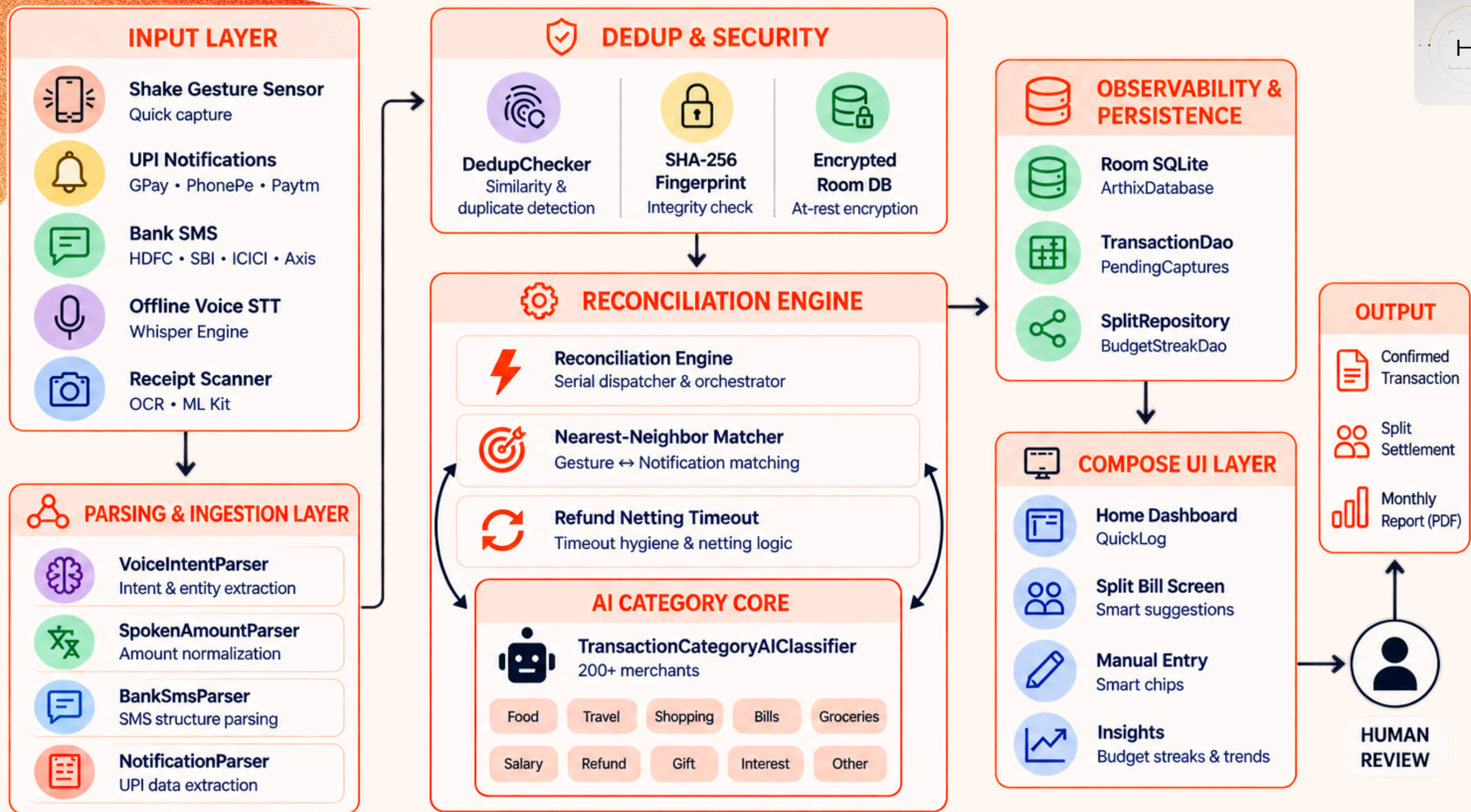
BUILT FOR INDIAN UPI USERS AND YOUNG PROFESSIONALS

- **Near-zero friction:** Capture spending in under 3 seconds, with zero typing, even during busy moments.
- **Four capture paths:** Shake, notification, voice, and OCR provide multiple ways to log a transaction.
- **Smart reconciliation:** Automatically matches user gestures with delayed bank notifications, preventing duplicate or missed records.
- **Frictionless bill splitting:** Suggests splits from historical spending patterns, reducing manual setup.
- **Grounded AI guidance:** Generates summaries, projections, and nudges from calculated financial data — never invented numbers.



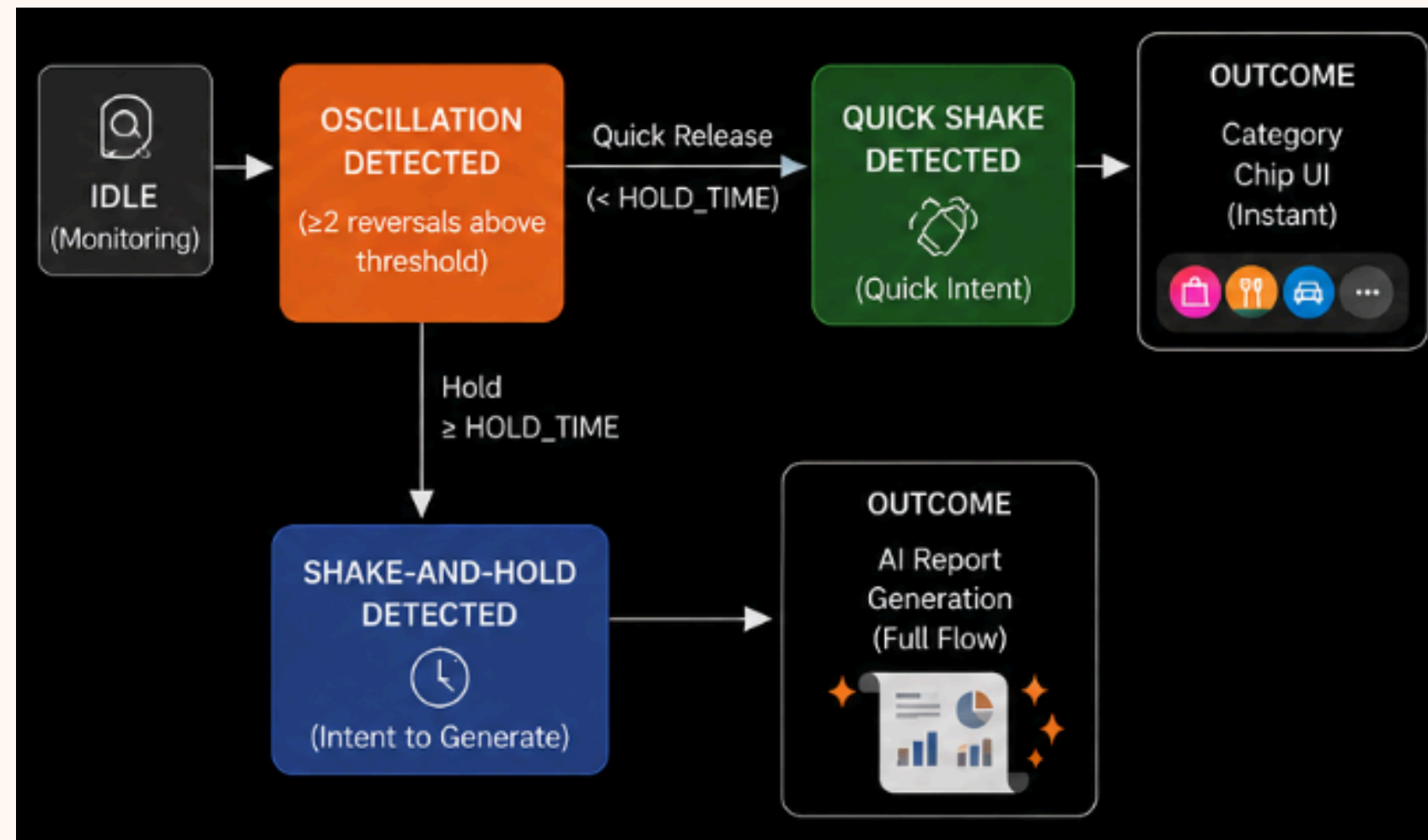
EVERY NUMBER ON SCREEN IS CALCULATED FIRST. THE LLM ONLY PHRASES IT — NEVER INVENTS IT.

SYSTEM ARCHITECTURE



SHAKE-TO-LOG & OSCILLATION DETECTION

HARDWARE-LEVEL GESTURE RECOGNITION THAT REJECTS FALSE POSITIVES



- Tuned accelerometer oscillation detector requires ≥ 2 directional reversals above threshold within a window
- Rejects false triggers from walking or pocket movement · Single state machine distinguishes a quick shake from a shake-and-hold gesture
- Quick shake → instant category chip UI · Shake-and-hold → triggers full AI report generation

FR-1 VS FR-7 DISAMBIGUATION LIVES IN ONE STATE MACHINE — NOT TWO COMPETING LISTENERS.

THE EVENT RECONCILIATION ENGINE

SOLVING THE RACE CONDITION BETWEEN GESTURE AND BANK NOTIFICATION

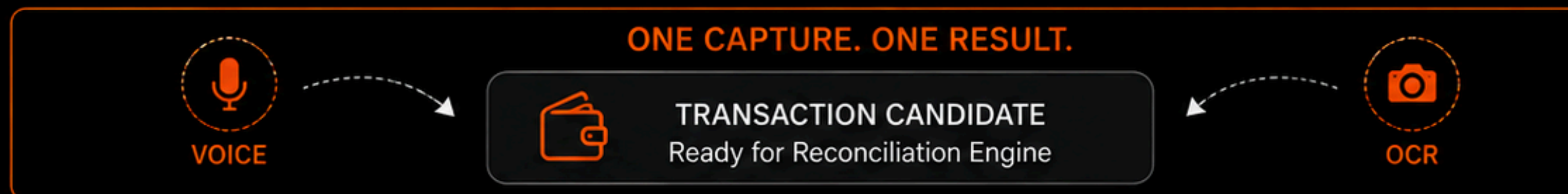
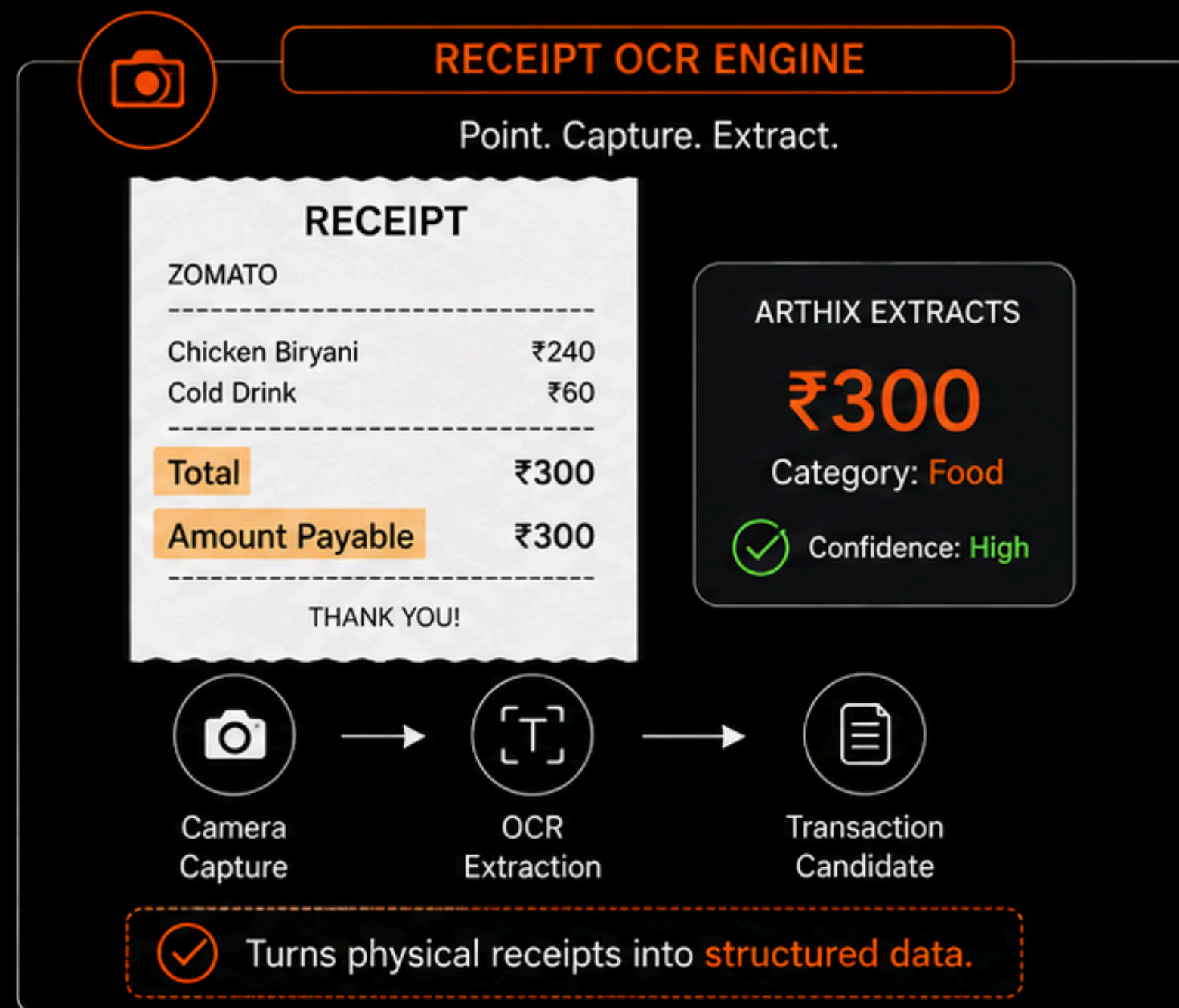
HIGHEST TECHNICAL DEPTH MODULE IN ARTHIX

- **SOLVES THE RACE CONDITION:** USER GESTURE AND BANK NOTIFICATION CAN ARRIVE AT DIFFERENT TIMES, OUT OF CHRONOLOGICAL ORDER
- **NEAREST-NEIGHBOR MONOTONIC TIME-MATCHING** USING `SYSTEMCLOCK.ELAPSEDREALTIME()` WITHIN A 2-MINUTE WINDOW
- **RUNS ON A DEDICATED SINGLE-THREADED SERIAL COROUTINE DISPATCHER**
- **SCORE-GAP DISAMBIGUATION:** WHEN TWO TRANSACTIONS CLUSTER IN TIME, THE SYSTEM NEVER SILENTLY GUESSES — IT SHOWS A ONE-TAP DISAMBIGUATION PROMPT
- **INDEPENDENT TIMEOUTS:** UNMATCHED SHAKES FALL BACK GRACEFULLY TO A PENDING QUEUE FOR VOICE FOLLOW-UP

NO SILENT GUESSING. IF THE SCORE GAP IS TOO SMALL, THE USER DECIDES — NOT THE ALGORITHM

VOICE & OCR EXPENSE CAPTURE

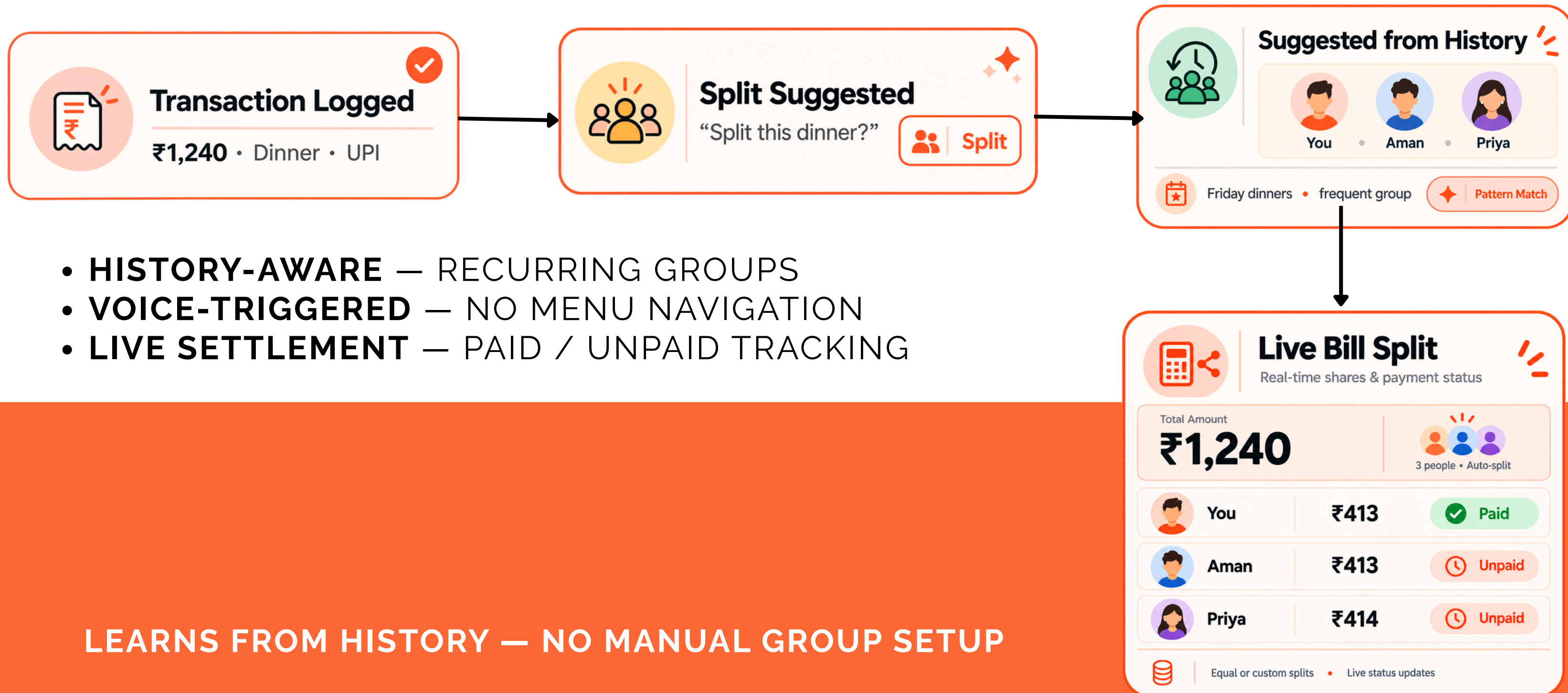
CAPTURE IN ANY SITUATION. ARTHIX STRUCTURES IT INSTANTLY.



SMART EXPENSE SPLITTING



CONTEXT-AWARE GROUP HEURISTICS WITH LIVE PAID-STATUS TRACKING

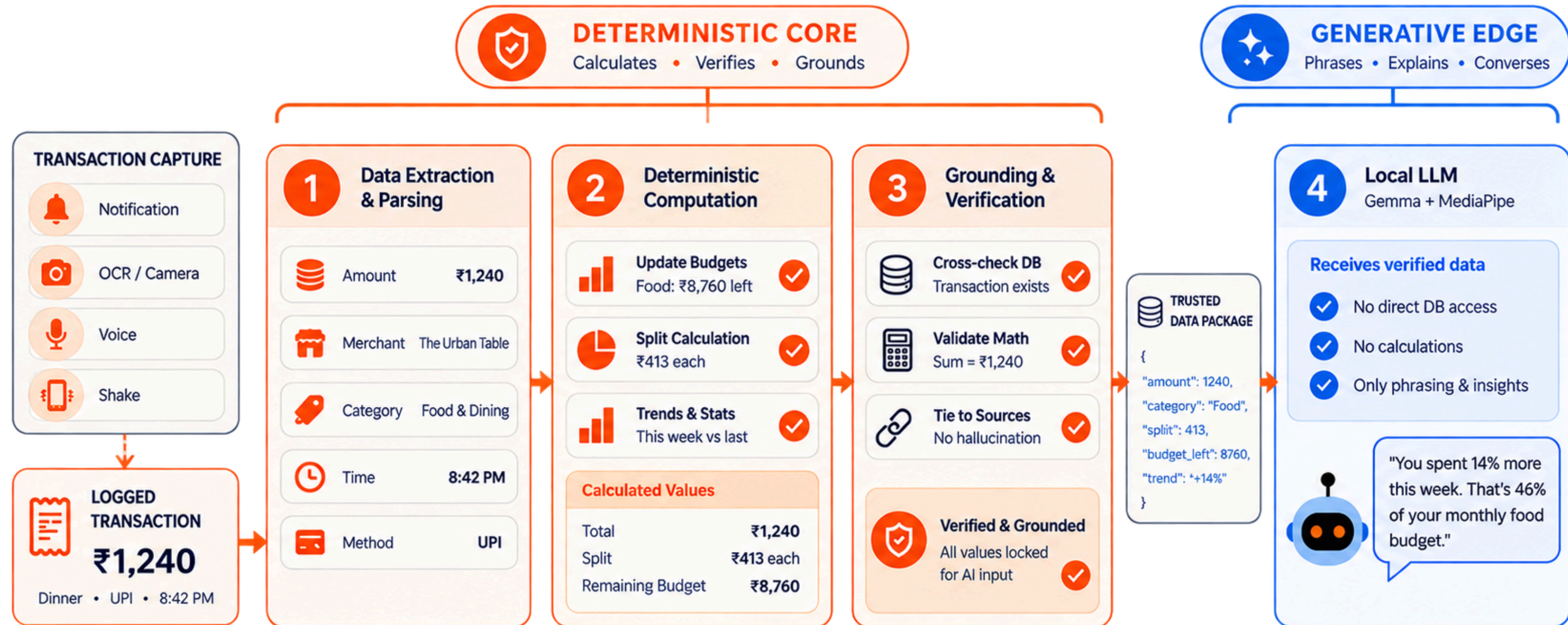


LEARNS FROM HISTORY — NO MANUAL GROUP SETUP

GROUNDING AI FINANCIAL INTELLIGENCE



DETERMINISTIC INTEGER MATH + A GROUNDED ON-DEVICE LLM



Every number on screen is **calculated first**.

The LLM only phrases it — **never invents it**.



INTEGER MATH

No floating-point money calculations



GROUNING SAFEGUARD

Output tied to verified values before it reaches the LLM



OFFLINE AI

Gemma runs locally through MediaPipe

PRIVACY, SECURITY & GAMIFICATION

LOCAL-FIRST ARCHITECTURE WITH HABIT-FORMING DESIGN



100% OFFLINE & PRIVATE

Zero cloud sync, zero data exfiltration,
local Room DB in WAL mode.



BIOMETRIC SECURITY

AppLock authentication gates
the entire app.



GAMIFIED LOGGING STREAKS

Daily logging habit tracked
and rewarded.



BUDGET ADHERENCE TRACKING

Visual feedback tied to streaks,
not just numbers.



TEST COVERAGE

175/175 unit tests passing,
zero failures.



**NOTHING ARTHIX
LOGS EVER
LEAVES THE DEVICE.**

LIVE DEMO & TECHNICAL METRICS

NUMBERS THAT BACK UP THE CLAIMS



LET'S BUILD SOMETHING THAT LASTS

From hackathon prototype to a habit-forming financial companion



ZERO FRICTION TODAY. FINANCIAL CLARITY FOR LIFE

1. **CONNECT** : COMPLETE FINANCIAL CONTEXT

- OPEN BANKING + ACCOUNT AGGREGATOR INTEGRATION
- AUTOMATIC BANK-DATA RECONCILIATION
- NO MANUAL IMPORTS OR CSV UPLOADS

2. **EXTEND**: CAPTURE EVERYWHERE

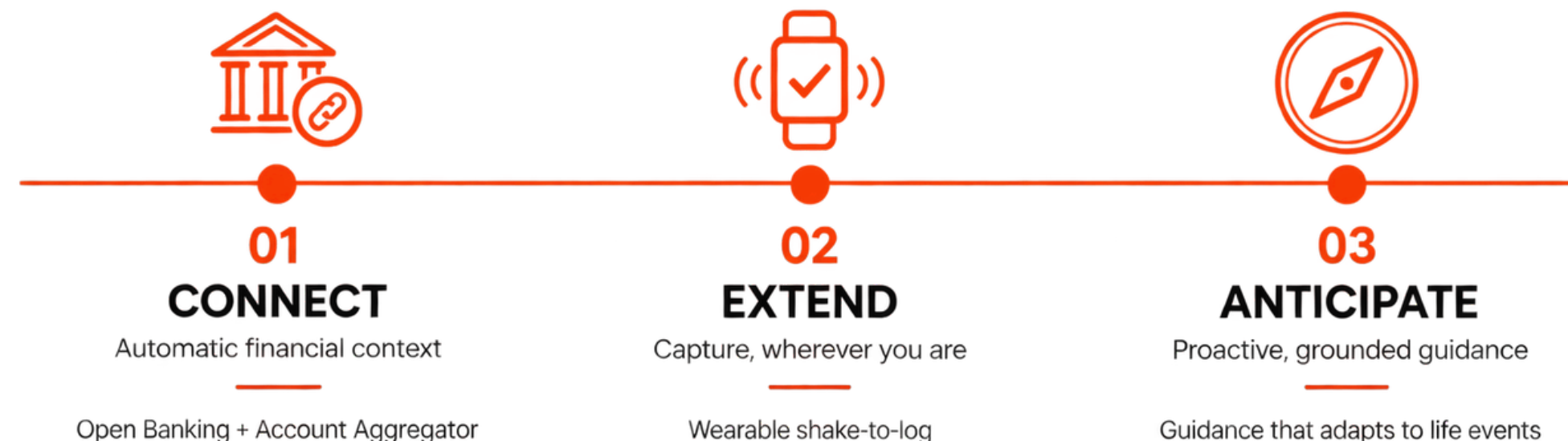
- WEARABLE GESTURE SUPPORT
- SHAKE-TO-LOG FROM A SMARTWATCH
- EXPENSE CAPTURE BEYOND THE PHONE

3. **ANTICIPATE**: FROM TRACKING TO GUIDANCE

- PROACTIVE SPENDING ALERTS
- GROUNDED RECOMMENDATIONS BEFORE OVERSPENDING
- GUIDANCE THAT ADAPTS TO LIFE EVENTS, NOT JUST PAST TRANSACTIONS

4. **THE LONG-TERM VISION**:

- ARTHIX BECOMES AN ALWAYS-ON FINANCIAL COMPANION
- UNDERSTANDS YOUR FINANCIAL CONTEXT
- ANTICIPATES NEEDS WHILE KEEPING FINANCIAL DECISIONS GROUNDED AND EXPLAINABLE



BETTER DATA



MORE NATURAL CAPTURE



MORE AUTONOMY

LET'S BUILD SOMETHING THAT LASTS

THANKYOU